

Unit 4

SYSTEMS WITH TWO DEGREE OF FREEDOM

Topics covered:

Mathematical modeling of two DOF system

Mathematical derivation of governing equations

(i) Newton's method

(ii) Energy method (Lagrange's method)

Solution to governing equations

Modes of Vibration

Co-ordinate coupling

Principal co-ordinates

Free vibration (linear, torsional, string and pendulum systems)

Forced Vibration

Applications

Introduction

In earlier classes we have discussed analysis of single degree of freedom systems. It is well known that *Degree of freedom* is the number of independent co-ordinates required to describe the motion of a vibratory system

Degree of freedom of a system = (Number of masses in a system) $\times$ (number of possible types of motion of each mass).

A two-degree freedom system is one that requires two coordinates to describe its motion. These coordinates are called generalized coordinates when they are independent of each other. A two DOF system has two equations of motion, which can be solved to obtain two natural frequencies

Examples of 2 DOF linear and torsional systems are shown in Fig.1.

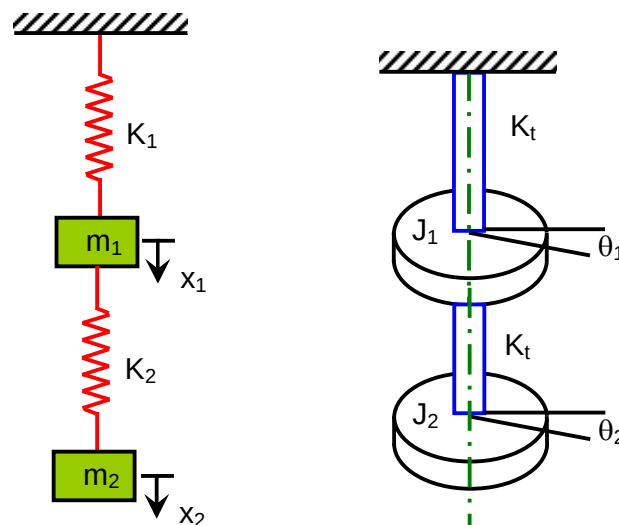


Fig.1 Examples of 2DOF linear and torsional systems

Obtain the equations of the motion of the linear system shown in Fig.1.

Governing equations can be obtained by

- (i) Newton's method
- (ii) Lagrange's method (Generalized energy method).

Newton's method

The Newton's method use force balance equations to obtain governing equations. The Force equilibrium diagram of given 2DOF linear system is shown in Fig.2.

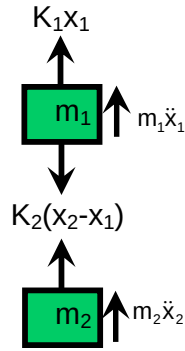


Fig.2 Force equilibrium diagram of 2DOF linear system

From Force equilibrium diagram of mass m1:

$$m_1\ddot{x}_1 + K_1x_1 - K_2(x_2 - x_1) = 0 \quad (1)$$

$$m_1\ddot{x}_1 + (K_1 + K_2)x_1 - K_2x_2 = 0 \quad (2)$$

From Force equilibrium diagram of mass m2 :

$$m_2\ddot{x}_2 + K_2(x_2 - x_1) = 0 \quad (3)$$

$$m_2\ddot{x}_2 - K_2x_1 + K_2x_2 = 0 \quad (4)$$

Eqns. (2) and (4) are Eqns of motion of the system.

Lagrange's method (Generalized energy method).

The Lagrange's equation is :

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_i} \right) - \frac{\partial T}{\partial x_i} + \frac{\partial U}{\partial x_i} = Q_i \quad (5)$$

where,

T - Kinetic energy of the system,

U - Potential energy of the system,

x_i - independent generalized displacement vector,

$\dot{x}_i$ -independent generalized velocity vector and

Q_i - generalized force vector.

$$T = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 \quad (6)$$

$$U = \frac{1}{2}K_1x_1^2 + \frac{1}{2}K_2(x_2 - x_1)^2 \quad (7)$$

$$\{x_i\} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

substitute Eqn.(6) and (7) in (5). The partial differentials are given below.

$$\frac{\partial T}{\partial \dot{x}_1} = m_1\dot{x}_1$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_1} \right) = m_1\ddot{x}_1$$

$$\frac{\partial T}{\partial x_1} = 0$$

$$\frac{\partial U}{\partial x_1} = K_1x_1 + K_2(x_2 - x_1)(-1)$$

$$\frac{\partial U}{\partial x_2} = x_1(K_1 + K_2) - K_2x_2$$

Substitute above differentials in Eqn. (5), to obtain the equations of motion.

$$m_1\ddot{x}_1 + (K_1 + K_2)x_1 - K_2x_2 = 0$$

$$m_2\ddot{x}_2 - K_2x_1 + K_2x_2 = 0$$

The above equations are similar to the equations obtained by Newton's method. So, one can use any one method to obtain the Eqns of motions.

Solution of governing equations

It is possible to have pure harmonic free vibration for both the masses.

Let us assume

$$x_1 = A_1\sin(\omega t + \phi) \quad (8)$$

$$x_2 = A_2\sin(\omega t + \phi) \quad (9)$$

The above equations have to satisfy the governing equations of motions. Substitute Eqns. (8) and (9) in (2) and (4).

$$[(K_1 + K_2) - m_1\omega^2]A_1\sin(\omega t + \phi) - K_2A_2\sin(\omega t + \phi) = 0 \quad (10)$$

$$-K_2A_1\sin(\omega t + \phi) - (K_2 - m_2\omega^2)A_2\sin(\omega t + \phi) = 0 \quad (11)$$

In above equations $\sin(\omega t + \phi) \neq 0$

The above equations reduces to: (characteristic equation)

$$[(K_1 + K_2) - m_1\omega^2]A_1 - K_2A_2 = 0 \quad (12)$$

$$-K_2A_1 - (K_2 - m_2\omega^2)A_2 = 0 \quad (13)$$

The solution of equations is:

$$\begin{vmatrix} (K_1 + K_2) - m_1\omega^2 & -K_2 \\ -K_2 & K_2 - m_2\omega^2 \end{vmatrix} = 0 \quad (14)$$

The above equation is referred as a characteristic determinant. Solving, we get :

$$m_1 m_2 \omega^4 - [m_2(K_1 + K_2) + m_1 K_2] \omega^2 + K_1 K_2 = 0 \quad (15)$$

$$\omega^4 - \left[\frac{(K_1 + K_2)}{m_1} + \frac{K_2}{m_2} \right] \omega^2 + \frac{K_1 K_2}{m_1 m_2} = 0 \quad (16)$$

For simplification, let us consider :

$$K_1 = K_2 = K$$

$$m_1 = m_2 = m$$

The frequency equation changes to:

$$\omega^4 - \left[\frac{2K}{m} + \frac{K}{m} \right] \omega^2 + \frac{K^2}{m^2} = 0 \quad (17)$$

$$\omega^4 - \left[\frac{3K}{m} \right] \omega^2 + \frac{K^2}{m^2} = 0 \quad (18)$$

put $\omega^2 = \lambda$, in the above equation:

$$\lambda^2 - \left[\frac{3K}{m} \right] \lambda + \frac{K^2}{m^2} = 0$$

$$\lambda_1, \lambda_2 = \omega_1^2, \omega_2^2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda_1, \lambda_2 = \omega_1^2, \omega_2^2 = \frac{\frac{3K}{m} \pm \sqrt{\left(\frac{3K}{m}\right)^2 - \frac{4K^2}{m^2}}}{2}$$

$$\omega_1^2, \omega_2^2 = \frac{(3 \pm \sqrt{5}) \frac{K}{m}}{2}$$

$$\omega_1^2, \omega_2^2 = \frac{\frac{3K}{m} \pm \sqrt{\frac{5K^2}{m^2}}}{2}$$

$$\omega_1 = \sqrt{\frac{(3 - \sqrt{5}) \frac{K}{m}}{2}} \quad \text{and} \quad \omega_2 = \sqrt{\frac{(3 + \sqrt{5}) \frac{K}{m}}{2}}$$

$$\omega_1 = 0.61\omega_n \quad \text{and} \quad \omega_2 = 1.61\omega_n \text{ radians/sec}$$

As the system has two natural frequencies, under certain conditions it may vibrate with first or second frequencies, which are referred as principal modes of vibration.

First principal mode of vibration

The system vibrates with first fundamental natural frequency, i.e $\omega = \omega_1$

For vibrations under Mode-I, consider.

A_{11} -amplitude of first mass (m_1) due to frequency ω_1

A_{21} -amplitude of second mass (m_2) due to frequency ω_1

Characteristic equations of the system changes to:

$$[2K - m\omega_1^2]A_{11} - KA_{21} = 0 \quad (19)$$

$$-KA_{11} - (K - m\omega_1^2)A_{21} = 0 \quad (20)$$

Let $\frac{A_{21}}{A_{11}} = \mu_1$ be amplitude ratio, from Eqn. (19) we get,

$$\mu_1 = \frac{A_{21}}{A_{11}} = \left(\frac{2K - m\omega_1^2}{K} \right) \quad (21)$$

Substitute ω_1 in the above equation.

$$\mu_1 = \frac{A_{21}}{A_{11}} = 2 - (0.61)^2 = 1.63 \quad (22)$$

First modal vector is:

$$\{A\}_1 = \begin{Bmatrix} A_{11} \\ A_{21} \end{Bmatrix} = \begin{Bmatrix} A_{11} \\ \mu_1 A_{11} \end{Bmatrix} = \begin{Bmatrix} 1 \\ 1.63 \end{Bmatrix} \quad (23)$$

Second principal mode of vibration

The system vibrates with first fundamental natural frequency, i.e $\omega = \omega_2$

For vibrations under Mode-II, consider

A_{12} -amplitude of first mass (m_1) due to frequency ω_2

A_{22} -amplitude of second mass (m_2) due to frequency ω_2

Characteristic equations of the system changes to:

$$[2K - m\omega_2^2]A_{12} - KA_{22} = 0 \quad (24)$$

$$-KA_{12} - (K - m\omega_2^2)A_{22} = 0 \quad (25)$$

Let $\frac{A_{22}}{A_{12}} = \mu_2$ be amplitude ratio, from Eqn. (24) we get,

$$[2K - m\omega_2^2]A_{12} - KA_{22} = 0 \quad (26)$$

$$\mu_2 = \frac{A_{22}}{A_{12}} = \left(\frac{2K - m\omega_2^2}{K} \right) \quad (27)$$

Substitute ω_2 in above equation,

$$\mu_2 = \frac{A_{22}}{A_{21}} = 2 - (1.61)^2 = -0.59 \quad (28)$$

Second modal vector is:

$$\{A\}_2 = \begin{Bmatrix} A_{12} \\ A_{22} \end{Bmatrix} = \begin{Bmatrix} A_{12} \\ \mu_2 A_{12} \end{Bmatrix} = \begin{Bmatrix} 1 \\ -0.59 \end{Bmatrix} \quad (29)$$

The modal matrix is:

$$[A] = [\{A\}_1 \{A\}_2] = \begin{bmatrix} A_{11} & A_{12} \\ \mu_1 A_{11} & \mu_2 A_{12} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1.63 & -0.59 \end{bmatrix} \quad (30)$$

Mode shapes of the system

The given vibratory system has two principal modes, mode shapes of the system are shown in Fig.3.

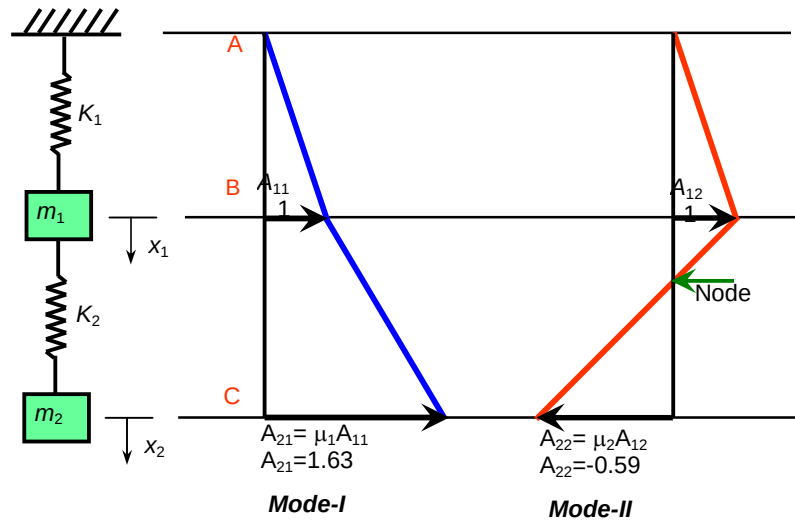


Fig.3 Mode shapes of the system

Torsional Vibratory systems

A torsional system is a rotor system as shown in Fig.4.

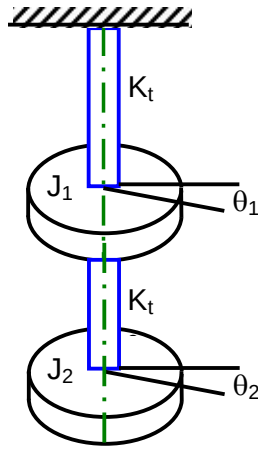


Fig.4 Examples of 2DOF torsional system

Obtain the equations of motion of the system shown in Fig.4.

Newton's method

The Newton's method use force balance equations to obtain governing equations. The Force equilibrium diagram of given 2DOF torsional system is shown in Fig.5.

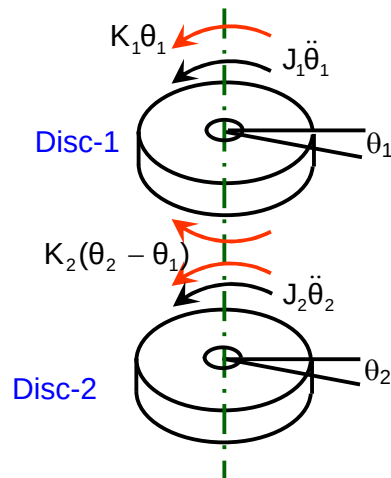


Fig.5 Force equilibrium diagram of 2DOF torsional system

From Force equilibrium diagram of disc 1

$$J_1 \ddot{\theta}_1 + K_1 \theta_1 - K_2 (\theta_2 - \theta_1) = 0$$

$$J_1 \ddot{\theta}_1 + (K_1 + K_2) \theta_1 - K_2 \theta_2 = 0$$

(31)

From Force equilibrium diagram of disc 2

$$J_2 \ddot{\theta}_2 + K_2 (\theta_2 - \theta_1) = 0$$

$$J_2 \ddot{\theta}_2 - K_2 \theta_1 + K_2 \theta_2 = 0 \quad (32)$$

Eqns.(31) and (32) are the governing Eqns. of the system.

Equivalent shaft for a torsional system

Many engineering applications we find stepped shaft as shown in Fig.6. For vibration it is required to have an equivalent system for the analysis. In this section how to obtain the torsionally equivalent shaft is illustrated.

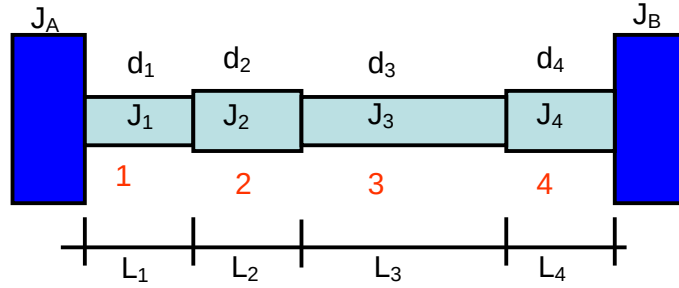


Fig.6. Stepped shaft

Let θ be the twist in the shaft by application of torque T , and θ_1 , θ_2 , θ_3 and θ_4 be twists in section 1, 2, 3 and 4 respectively.

$$\theta = \theta_1 + \theta_2 + \theta_3 + \theta_4 \quad (33)$$

From theory of torsion,

$$\frac{T}{J} = \frac{G\theta}{L} \quad (34)$$

Where,

$$J = \frac{\pi d^4}{32} \quad (35)$$

$$\theta = \frac{TL_1}{J_1 G_1} + \frac{TL_2}{J_2 G_2} + \frac{TL_3}{J_3 G_3} + \frac{TL_4}{J_4 G_4} \quad (36)$$

If material of shaft is same, then above equation can be written as:

$$\theta = \frac{32}{\pi G} T \left(\frac{L_1}{d_1^4} + \frac{L_2}{d_2^4} + \frac{L_3}{d_3^4} + \frac{L_4}{d_4^4} \right) \quad (37)$$

If d_e and L_e are equivalent diameter and lengths of the shaft, then:

$$\frac{L_e}{d_e^4} = \left(\frac{L_1}{d_1^4} + \frac{L_2}{d_2^4} + \frac{L_3}{d_3^4} + \frac{L_4}{d_4^4} \right) \quad (38)$$

$$L_e = L_1 \left(\frac{d_e}{d_1} \right)^4 + L_2 \left(\frac{d_e}{d_2} \right)^4 + L_3 \left(\frac{d_e}{d_3} \right)^4 + L_4 \left(\frac{d_e}{d_4} \right)^4 \quad (39)$$

The equivalent shaft of a system shown in Fig.6 given by Eqn.(39) is shown in Fig.7.

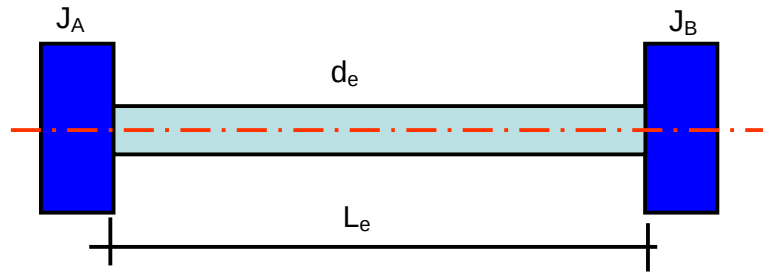


Fig.7. Equivalent shaft of the system shown in Fig.6

Definite and semi-definite systems

A system, which is free from both the ends, is referred as semi-definite system.

Ex: Train, Turbo-generator, etc.

For semi-definite systems, the first natural frequency is zero. A simple semi-definite linear and a torsional systems are shown in Fig.8.

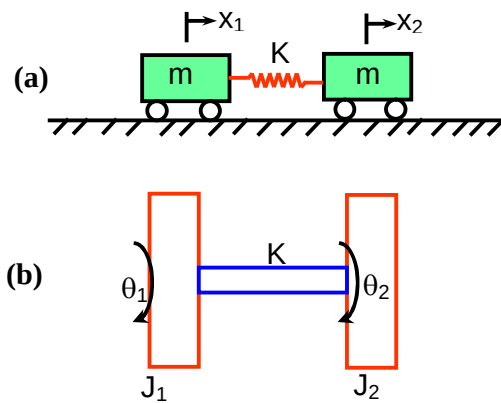


Fig.8. Semi-definite systems (a) linear and (b) torsional.

A system, which is fixed from one end or both the ends is referred as definite system. A definite system has nonzero lower natural frequency. Various definite linear and a torsional systems are shown in Fig.9.

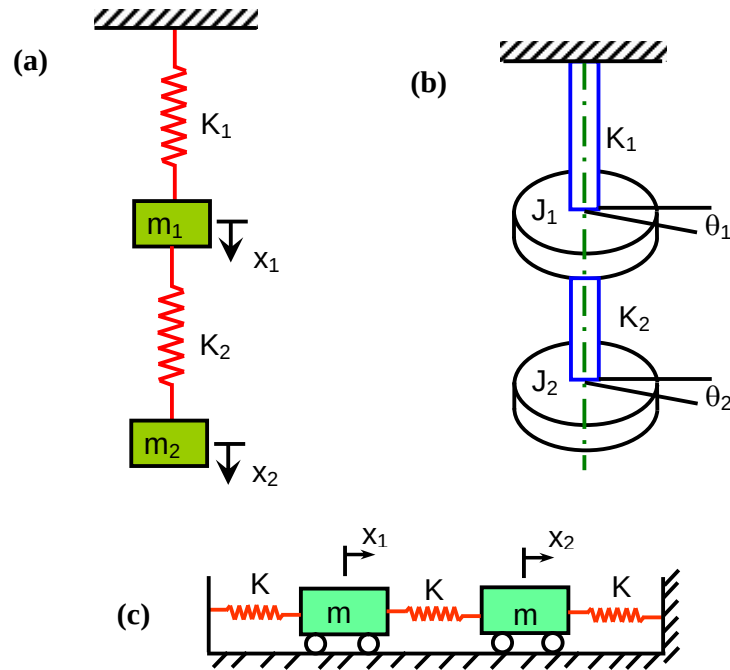


Fig.9. Various definite systems

Example 1

For the torsional system shown in the Fig.10,

- (i) derive the equations of motion,
- (ii) setup the frequency equation and obtain the fundamental natural frequencies
- (iii) obtain the modal vectors and modal matrix
- (iv) draw mode shapes of the system.

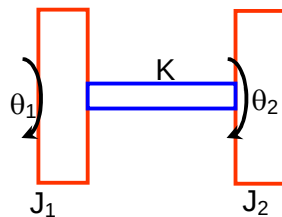


Fig.10. Turbo-generator

Solution:

Write the Equations of motions of the system, use Lagrange's equation.

The Lagrange's equation is :

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_i} \right) - \frac{\partial T}{\partial \theta_i} + \frac{\partial U}{\partial \theta_i} = Q_i \quad (40)$$

where,

$$T = \frac{1}{2} J_1 \dot{\theta}_1^2 + \frac{1}{2} J_2 \dot{\theta}_2^2 \quad (41)$$

$$U = \frac{1}{2} K_2 (\theta_2 - \theta_1)^2 \quad (42)$$

$$\{\theta_i\} = \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix}$$

substitute Eqn. (41) and (42) in Eqn.(40) to get the equations of motion:

$$J_1 \ddot{\theta}_1 + K \theta_1 - K \theta_2 = 0 \quad (43)$$

$$J_2 \ddot{\theta}_2 - K \theta_1 + K \theta_2 = 0 \quad (44)$$

Assuming the vibration of the system is SHM,

$$\theta_1 = A_1 \sin(\omega t + \phi) \quad (45)$$

$$\theta_2 = A_2 \sin(\omega t + \phi) \quad (46)$$

The above equations have to satisfy the governing equations of motions Eqn. (43) and (44). Simplifying, we get.

$$[K - J_1 \omega^2] A_1 \sin(\omega t + \phi) - K A_2 \sin(\omega t + \phi) = 0 \quad (47)$$

$$-K A_1 \sin(\omega t + \phi) - (K - J_2 \omega^2) A_2 \sin(\omega t + \phi) = 0 \quad (48)$$

In above equations

$$\sin(\omega t + \phi) \neq 0$$

The above equations reduces to: (characteristic equation)

$$[K - J_1 \omega^2] A_1 - K A_2 = 0 \quad (49)$$

$$-K A_1 - (K - J_2 \omega^2) A_2 = 0 \quad (50)$$

The above equations can be written in the determinant form as:

$$\begin{vmatrix} K - J_1 \omega^2 & -K \\ -K & K - J_2 \omega^2 \end{vmatrix} = 0 \quad (51)$$

The above equation is referred as a characteristic determinant. Solving, we get the determinant,

$$J_1 J_2 \omega^4 - [J_1 K + J_2 K] \omega^2 = 0 \quad (52)$$

$$\omega^4 - \left[\frac{K}{J_1} + \frac{K}{J_2} \right] \omega^2 = 0 \quad (53)$$

The above equation is referred as frequency equation. Solving the frequency equations, we get.

$$\omega_1 = 0 \text{ rad/s and } \omega_2 = \sqrt{K \left(\frac{J_1 + J_2}{J_1 J_2} \right)} \text{ rad/s.}$$

For simplification, let us consider,

$$J_1 = J_2 = J, \quad (54)$$

$$\text{then } \omega_1 = 0 \text{ rad/s and } \omega_2 = \sqrt{\frac{2K}{J}} \text{ rad/s}$$

As the system is semi-definite system, the first natural frequency is found to be ZERO. The system has two natural frequencies. Under certain conditions it may vibrate with first or second frequency, which are referred as principal modes of vibration.

Now, the characteristic equations of the system due to condition in Eqn.(54) are:

$$[K - J_1 \omega^2] A_1 - K A_2 = 0 \quad (55)$$

$$-K A_1 - (K - J_2 \omega^2) A_2 = 0 \quad (56)$$

First principal mode of vibration

The system vibrates with first fundamental natural frequency, i.e ω_1 .

For vibrations under Mode-I, consider,

A_{11} -amplitude of first disc (J_1) due to frequency ω_1

A_{21} -amplitude of second disc (J_2) due to frequency ω_1

Characteristic equations of the system changes to:

$$[K - J_1 \omega^2] A_{11} - K A_{21} = 0 \quad (57)$$

$$-K A_{11} - (K - J_2 \omega^2) A_{21} = 0 \quad (58)$$

Let, $\frac{A_{21}}{A_{11}} = \mu_1$ be amplitude ratio

Then, from Eqn. (57) one can obtain,

$$\mu_1 = \frac{A_{21}}{A_{11}} = \left(\frac{K - J \omega_1^2}{K} \right) \quad (59)$$

Substitute ω_1 in above equation.

$$\mu_1 = \frac{A_{21}}{A_{11}} = 1$$

First modal vector is:

$$\{A\}_1 = \begin{Bmatrix} A_{11} \\ A_{21} \end{Bmatrix} = \begin{Bmatrix} A_{11} \\ \mu_1 A_{11} \end{Bmatrix} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \quad (60)$$

Second principal mode of vibration

For vibrations under Mode-II, consider,

A_{12} -amplitude of first disc (J_1) due to frequency ω_2

A_{22} -amplitude of second disc (J_2) due to frequency ω_2

Characteristic equations of the system changes to:

$$[K - J_1\omega^2]A_{12} - KA_{22} = 0 \quad (61)$$

$$-KA_{12} - (K - J_2\omega^2)A_{22} = 0 \quad (62)$$

Let $\frac{A_{22}}{A_{12}} = \mu_2$ is the amplitude ratio,

From Eqn. (61) on can obtain,

$$\mu_2 = \frac{A_{21}}{A_{11}} = \left(\frac{K - J\omega_2^2}{K} \right)$$

Substitute ω_2 in the above equation,

$$\mu_1 = \frac{A_{21}}{A_{11}} = \left(\frac{K - J\frac{2K}{J}}{K} \right) = -1 \quad (63)$$

Second modal vector is:

$$\{A\}_2 = \begin{Bmatrix} A_{12} \\ A_{22} \end{Bmatrix} = \begin{Bmatrix} A_{12} \\ \mu_2 A_{12} \end{Bmatrix} = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \quad (64)$$

Modal matrix

$$[A] = [\{A\}_1 \{A\}_2] = \begin{bmatrix} A_{11} & A_{12} \\ \mu_1 A_{11} & \mu_2 A_{12} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (65)$$

Mode shapes of the system

The mode shapes obtained from modal matrix Eqn. (65) are shown in Fig.11.

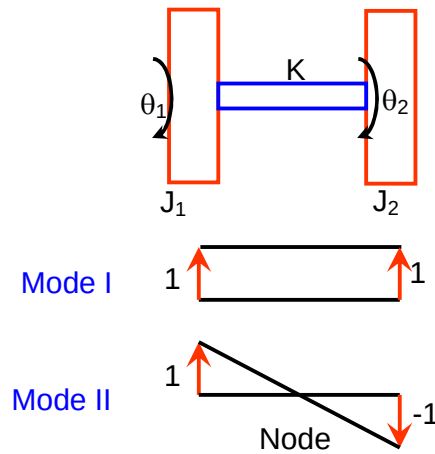


Fig.11. Mode shapes of the system

Various geared systems

A Turbo-generator geared system is shown in the fig.12. The analysis of this system is complex due to the gears. The system can be made simple as discussed below.

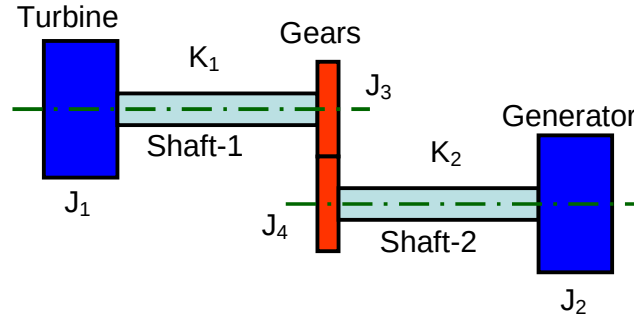


Fig.12 Turbo-generator geared system.

The speed ratio, n of the system is:

$$n = \frac{\text{speed of turbine}}{\text{speed of generator}}$$

For the system given in Fig. 12, the simplification can be made by,

(i) considering inertia of gears and (ii) neglecting inertia of gears.

Inertia of gears is considered

The system becomes 3 DOF system as shown in Fig.13.

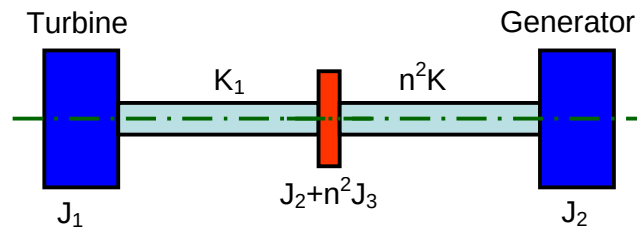


Fig.13 Turbo-generator geared system considering the inertia of gears

Inertia of gears is neglected

The system becomes 2 DOF system as shown in Fig.14.

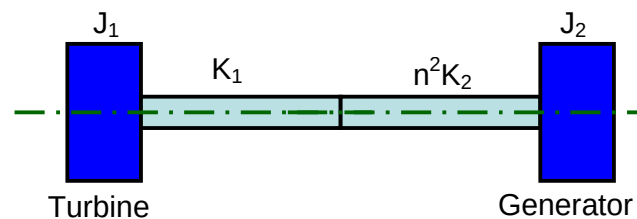


Fig.14 Turbo-generator geared system neglecting the inertia of gears

Two torsional springs are in series, therefore the effective stiffness can be obtained from,

$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{n^2 K_2} \quad (66)$$

Practice problems (Linear definite systems)

For the systems shown in the Fig.15,

- (i) derive the equations of motion,
- (ii) setup the frequency equation and obtain the fundamental natural frequencies
- (iii) obtain the modal vectors and modal matrix
- (iv) draw mode shapes of the system.

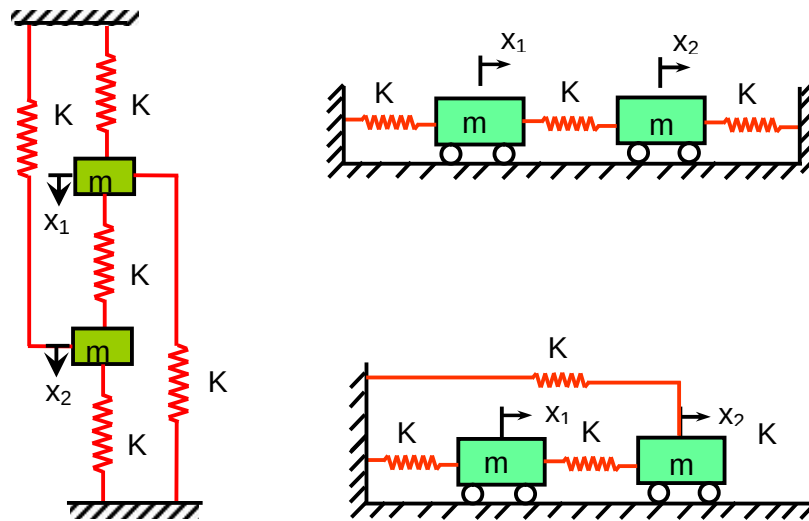


Fig.15 Linear vibratory systems

Practice problems (Torsional semi-definite systems)

For the systems shown in figure, obtain natural frequencies and mode shapes

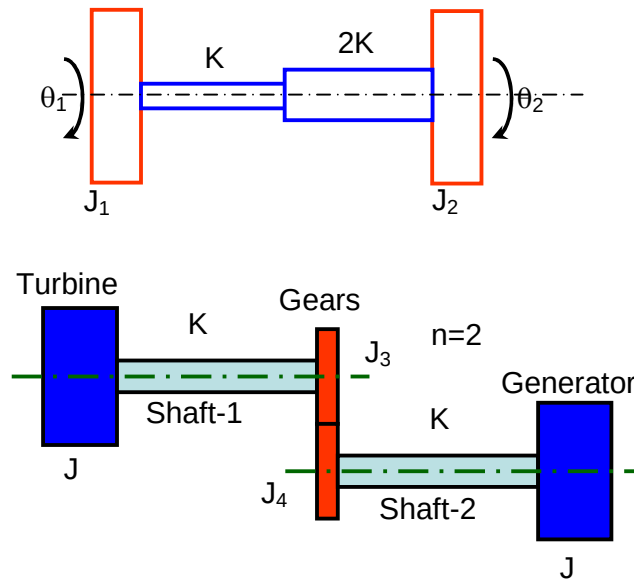


Fig.16 Torsional vibratory system.

Pendulum systems (double pendulum)

Estimate the natural frequencies of the double pendulum shown in the Fig.17. For simplicity take $m_1=m_2=m$ and $l_1=l_2=l$.

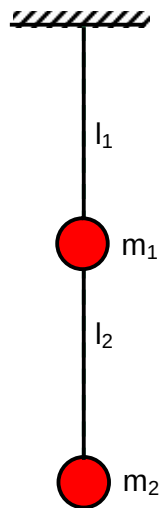


Fig.17 Double pendulum.

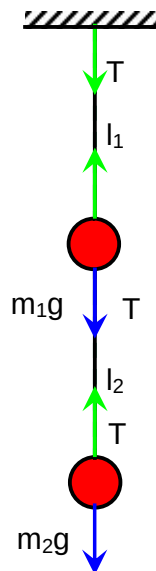


Fig.18 Static forces in Double pendulum.

Due to self weight of masses, the pendulum rods are in tension, the static forces in the double pendulum is shown in the Fig.18. From this figure one can write,

$$T_2 = m_2 g \quad (67)$$

$$T_1 = m_1 g + T_2 \quad (68)$$

$$T_1 = m_1 g + m_2 g = (m_1 + m_2) g \quad (69)$$

Above equations holds good for small oscillations.

When the system is disturbed, it starts vibrating, a position at any instant of time with forces is shown ion Fig.19.

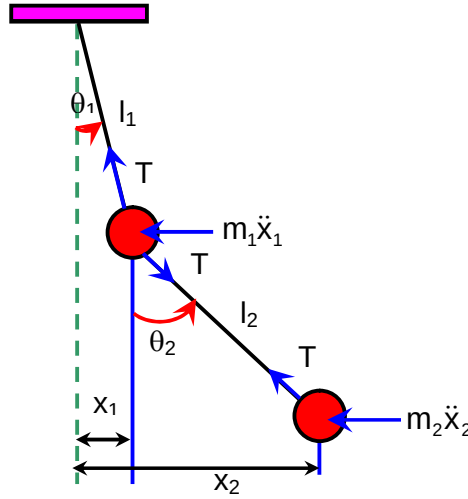


Fig.19 Dynamic forces on Double pendulum

Consider x_1 and x_2 are generalized co-ordinates. From Fig.19, the relation between x and θ can be written as:

$$\sin\theta_1 = \theta_1 = \frac{x_1}{l_1} \quad (70)$$

$$\sin\theta_2 = \theta_2 = \frac{x_2 - x_1}{l_2} \quad (71)$$

Resolving the forces on the mass m_1 , (Ref. Fig.19) we get,

$$m_1 \ddot{x}_1 + T_1 \theta_1 - T_2 \theta_2 = 0 \quad (72)$$

Substitute, Eqns. (67), (69), (70), and (71) in eqns. (72).

$$m_1 \ddot{x}_1 + T_1 \frac{x_1}{l_1} - T_2 \frac{x_2 - x_1}{l_2} = 0 \quad (73)$$

$$m_1 \ddot{x}_1 + (m_1 + m_2) g \frac{x_1}{l_1} - (m_2 g) \frac{x_2 - x_1}{l_2} = 0 \quad (74)$$

Substitute $m_1 = m_2 = m$ and $l_1 = l_2 = l$ in the above Eqn.

$$m \ddot{x}_1 + (2m) g \frac{x_1}{l} - (mg) \frac{x_2 - x_1}{l} = 0 \quad (75)$$

$$m\ddot{x}_1 + \left(\frac{3mg}{l}\right)x_1 - \left(\frac{mg}{l}\right)x_2 = 0 \quad (76)$$

Above Eqn. is first Eqn. of motion.

Resolving the forces on the mass m_2 , (Ref. Fig.19) we get,

$$m_2\ddot{x}_2 + T_2\theta_2 = 0 \quad (77)$$

Substitute, Eqns. (67), (69), (70), and (71) in eqns. (77).

$$m_2\ddot{x}_2 + T_2 \frac{x_2 - x_1}{l_2} = 0 \quad (78)$$

$$m_2\ddot{x}_2 + m_2g\left(\frac{x_2 - x_1}{l_2}\right) = 0 \quad (79)$$

Substitute $m_1=m_2=m$ and $l_1=l_2=l$ in the above Eqn.

$$m\ddot{x}_2 - \left(\frac{mg}{l}\right)x_1 + \left(\frac{mg}{l}\right)x_2 = 0 \quad (80)$$

Above Eqn. is second Eqn. of motion.

Eqns.(76) and (80) are governing equations of motion of the system.

Solution to governing eqns.:

Solution to governing eqns. can be obtained by considering the response of each mass as SHM:

$$x_1 = A_1\sin(\omega t + \phi) \quad (81)$$

$$x_2 = A_2\sin(\omega t + \phi) \quad (82)$$

The above equations have to satisfy the governing equations of motions Eqns.(76) and (80). Substitute eqns. (81) and (82) in Eqns.(76) and (80).

$$\left[\frac{3mg}{l} - m\omega^2\right]A_1\sin(\omega + \phi) - \frac{mg}{l}A_2\sin(\omega + \phi) = 0 \quad (83)$$

$$-\frac{mg}{l}A_1\sin(\omega + \phi) - \left[\frac{mg}{l} - m\omega^2\right]A_2\sin(\omega + \phi) = 0 \quad (84)$$

In above equations $\sin(\omega t + \phi) \neq 0$

Therefore, the above equations reduces to: (characteristic equation)

$$\left[\frac{3mg}{l} - m\omega^2\right]A_1 - \frac{mg}{l}A_2 = 0 \quad (85)$$

$$-\frac{mg}{l}A_1 - \left[\frac{mg}{l} - m\omega^2\right]A_2 = 0 \quad (86)$$

The above Eqns can be written in determinant form as:

$$\begin{vmatrix} \frac{3mg}{l} - m\omega^2 & -\frac{mg}{l} \\ -\frac{mg}{l} & \frac{mg}{l} - m\omega^2 \end{vmatrix} = 0 \quad (87)$$

The above equation is referred as a characteristic determinant, Solving, we get :

$$m^2\omega^4 - \frac{4m^2\omega^2g}{l} + \frac{2m^2g^2}{l^2} = 0 \quad (88)$$

The above equation is referred as a frequency equation

Solving Eqn. (88), we get:

Natural frequencies of the system

$$\omega_1 = 0.27\sqrt{\frac{g}{l}} \text{ radians/s} \quad \omega_2 = 3.73\sqrt{\frac{g}{l}} \text{ radians/s}$$

Modal vectors and the mode shapes can be obtained by usual manner, as explained earlier.

String systems

Obtain the natural frequencies of the string system shown in the Fig.20. For simplicity take $m_1=m_2=m$ and $l_1=l_2=l_3=l$

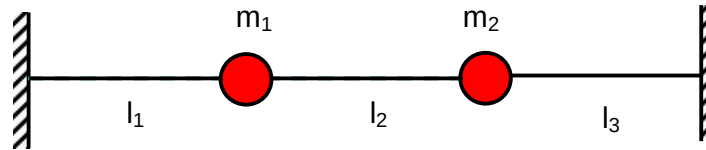


Fig.20 String system

For the above system, small angular oscillations, it can be assumed that the tensions in the string (T) do not change. The free body diagram of the system at any instant of time is shown in Fig.21.

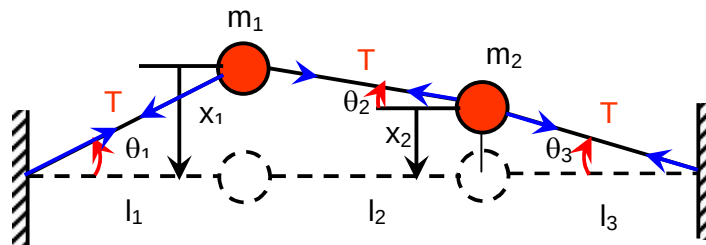


Fig.21 Free body diagram of the system at time, t

Consider x_1, x_2 as generalized co-ordinates, from Fig.21, the relations between θ and x are:

$$\theta_1 = \frac{x_1}{l}$$

$$\theta_2 = \frac{x_1 - x_2}{l}$$

$$\theta_3 = \frac{x_2}{l}$$

Considering the force equilibrium of first mass, m_1

$$m\ddot{x}_1 + T\theta_1 + T\theta_2 = 0 \quad (89)$$

Substitute for angular motions $\theta_1, \theta_2, \theta_3$ from above eqns.

$$\begin{aligned} m\ddot{x}_1 + T \frac{x_1}{l} + T \frac{x_1 - x_2}{l} &= 0 \\ m\ddot{x}_1 + \left(\frac{2T}{l}\right)x_1 - \left(\frac{T}{l}\right)x_2 &= 0 \end{aligned} \quad (90)$$

Considering the force equilibrium of second mass, m_2

$$m\ddot{x}_2 - T\theta_2 + T\theta_3 = 0 \quad (91)$$

Substitute for angular motions $\theta_1, \theta_2, \theta_3$ from above eqns.

$$\begin{aligned} m\ddot{x}_2 - T \frac{x_1 - x_2}{l} + T \frac{x_2}{l} &= 0 \\ m\ddot{x}_2 - \left(\frac{T}{l}\right)x_1 + \left(\frac{2T}{l}\right)x_2 &= 0 \end{aligned} \quad (92)$$

Eqns. (90) and (92) are two equations of motion of the system.

Solution to governing eqns.:

Solution to governing eqns. can be obtained by considering the response of each mass as SHM:

$$x_1 = A_1 \sin(\omega t + \phi) \quad (92)$$

$$x_2 = A_2 \sin(\omega t + \phi) \quad (93)$$

The above equations have to satisfy the governing equations of motions Eqns.(90) and (92). Substitute eqns. (92) and (93) in Eqns.(90) and (92).

$$\left[\frac{2T}{l} - m\omega^2 \right] A_1 \sin(\omega t + \phi) - \frac{T}{l} A_2 \sin(\omega t + \phi) = 0 \quad (94)$$

$$-\frac{T}{l} A_1 \sin(\omega t + \phi) + \left[\frac{2T}{l} - m\omega^2 \right] A_2 \sin(\omega t + \phi) = 0 \quad (95)$$

In above equations $\sin(\omega t + \phi) \neq 0$

Therefore, the above equations reduces to: (characteristic equation)

$$\left[\frac{2T}{l} - m\omega^2 \right] A_1 - \frac{T}{l} A_2 = 0 \quad (96)$$

$$-\frac{T}{l} A_1 - \left[\frac{2T}{l} - m\omega^2 \right] A_2 = 0 \quad (97)$$

The above Eqns can be written in determinant form as:

$$\begin{vmatrix} \frac{2T}{l} - m\omega^2 & -\frac{T}{l} \\ -\frac{T}{l} & \frac{2T}{l} - m\omega^2 \end{vmatrix} = 0 \quad (98)$$

The above equation is referred as a characteristic determinant, Solving, we get :

$$m^2\omega^4 - \frac{4Tm\omega^2}{l} + \frac{3T^2}{l^2} = 0 \quad (99)$$

The above equation is referred as a frequency equation. Solving Eqn. (99), we get:

Natural frequencies of the system

$$\omega_1 = \sqrt{\frac{T}{ml}} \text{ radians/s} \quad \omega_2 = \sqrt{\frac{3T}{ml}} \text{ radians/s}$$

As the system has two natural frequencies, under certain conditions it may vibrate with first or second frequency, which is referred as principal modes of vibration.

First principal mode of vibration

The system vibrates with first fundamental natural frequency, i.e. ω_1 .

For vibrations under Mode-I, consider,

A_{11} -amplitude of first mass (m_1) due to frequency ω_1

A_{21} -amplitude of second mass (m_2) due to frequency ω_1

The characteristic Eqns.(96) and (97) of the system changes to:

$$\left[\frac{2T}{l} - m\omega_1^2 \right] A_{11} - \frac{T}{l} A_{21} = 0 \quad (100)$$

$$-\frac{T}{l} A_{11} - \left[\frac{2T}{l} - m\omega_1^2 \right] A_{21} = 0 \quad (101)$$

Let $\frac{A_{21}}{A_{11}} = \mu_1$ is amplitude ratio. To compute μ_1 one can use any of the above eqns.

Consider Eqn.(100), we get μ_1 as:

$$\mu_1 = \frac{A_{21}}{A_{11}} = \left(\frac{\frac{2T}{l} - m\omega_1^2}{\frac{T}{l}} \right) \quad (102)$$

Substitute ω_1 in the above Eqn.

$$\mu_1 = \frac{A_{21}}{A_{11}} = \left(\frac{\frac{2T}{l} - m \frac{T}{ml}}{\frac{T}{l}} \right) = 1 \quad (103)$$

First modal vector is:

$$\{A\}_1 = \begin{Bmatrix} A_{11} \\ A_{21} \end{Bmatrix} = \begin{Bmatrix} A_{11} \\ \mu_1 A_{11} \end{Bmatrix} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \quad (104)$$

Second principal mode of vibration

The system vibrates with first fundamental natural frequency, i.e. ω_2 .

For vibrations under Mode-II, consider,

A_{12} -amplitude of first mass (m_1) due to frequency ω_2

A_{22} -amplitude of second mass (m_2) due to frequency ω_2

The characteristic Eqns.(96) and (97) of the system changes to:

$$\left[\frac{2T}{l} - m\omega_2^2 \right] A_{12} - \frac{T}{l} A_{22} = 0 \quad (105)$$

$$- \frac{T}{l} A_{12} - \left[\frac{2T}{l} - m\omega_2^2 \right] A_{22} = 0 \quad (106)$$

Let $\frac{A_{22}}{A_{12}} = \mu_2$ is amplitude ratio. To compute μ_2 one can use any of the above eqns.

Consider Eqn.(105), we get μ_2 as:

$$\mu_2 = \frac{A_{22}}{A_{12}} = \left(\frac{\frac{2T}{l} - m\omega_2^2}{\frac{T}{l}} \right) \quad (107)$$

Substitute ω_2 in the above Eqn.

$$\mu_2 = \frac{A_{21}}{A_{11}} = \left(\frac{\frac{2T}{l} - m \frac{3T}{ml}}{\frac{T}{l}} \right) = 1 \quad (108)$$

Second modal vector is:

$$\{A\}_2 = \begin{Bmatrix} A_{12} \\ A_{22} \end{Bmatrix} = \begin{Bmatrix} A_{12} \\ \mu_2 A_{12} \end{Bmatrix} = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \quad (109)$$

Modal matrix

$$[A] = [\{A\}_1 \{A\}_2] = \begin{bmatrix} A_{11} & A_{12} \\ \mu_1 A_{11} & \mu_2 A_{12} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (110)$$

Mode shapes of the system are shown in Fig.22.

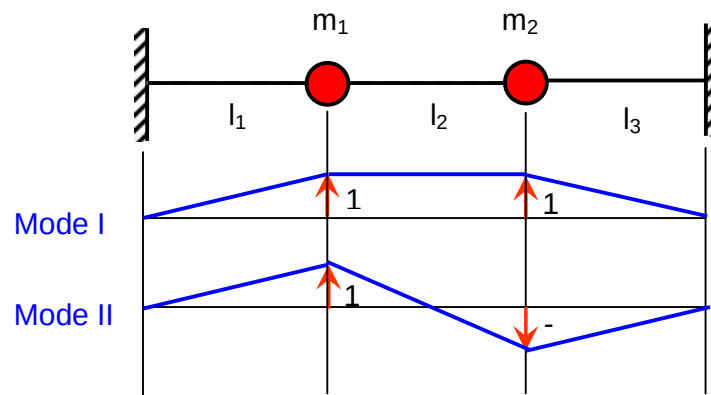


Fig.22. Mode shapes of the system

CO-ORDINATE COUPLING AND PRINCIPAL COORDINATES

Example-1

Obtain the equations of motion and natural frequencies of the system shown in the Fig.23. The vibration is restricted in plane of paper.

m -mass of the system

J -mass MI of the system

G -centre of gravity

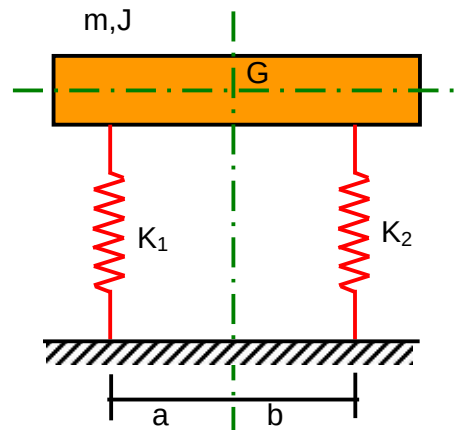


Fig.23. A 2 dof vibratory system

The system has two generalized co-ordinates, x and θ . x is in Cartesian and θ is in Polar co-ordinate systems. For small disturbance the system moves as shown in Fig.24(a). The amount displacements in two springs are shown in Fig.24(b).

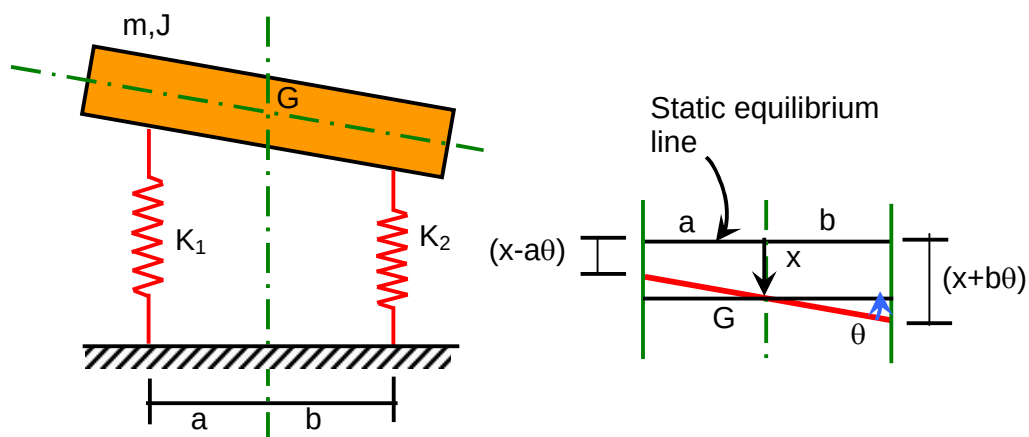


Fig.24 (a) system under vibration (b) displacements in springs at point 1 and 2

The Lagrange's equation is :

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_i} \right) - \frac{\partial T}{\partial x_i} + \frac{\partial U}{\partial x_i} = Q_i \quad (111)$$

generalized co-ordinates are:

$$\{x_i\} = \begin{Bmatrix} x \\ \theta \end{Bmatrix}$$

Kinetic energy of the system is:

$$T = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} J \dot{\theta}^2 \quad (112)$$

Potential energy of the system is:

$$U = \frac{1}{2} K_1 (x - a\theta)^2 + \frac{1}{2} K_2 (x + b\theta)^2 \quad (113)$$

First Lagrange's Eqn is:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) - \frac{\partial T}{\partial x} + \frac{\partial U}{\partial x} = Q_1 \quad (114)$$

Substitute Eqn. (112) and (113) in (114)

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) &= m \ddot{x} \\ \frac{\partial T}{\partial x} &= 0 \\ \frac{\partial U}{\partial x} &= K_1 (x - a\theta)(1) + K_2 (x + b\theta)(1) \end{aligned}$$

1st Equation of motion is:

$$\begin{aligned} \frac{\partial U}{\partial x} &= K_1 (x - a\theta)(1) + K_2 (x + b\theta)(1) \\ \frac{\partial U}{\partial x} &= K_1 x - K_1 a\theta + K_2 x + K_2 b\theta \\ \frac{\partial U}{\partial x} &= x(K_1 + K_2) - \theta(K_1 a - K_2 b) \\ m \ddot{x} + (K_1 + K_2)x - (K_1 a - K_2 b)\theta &= 0 \end{aligned} \quad (115)$$

Second Lagrange's Eqn is:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}} \right) - \frac{\partial T}{\partial \theta} + \frac{\partial U}{\partial \theta} = Q_2 \quad (116)$$

Substitute Eqn. (112) and (113) in (116)

$$J \ddot{\theta} - (K_1 a - K_2 b)x + (K_1 a^2 + K_2 b^2)\theta = 0 \quad (117)$$

eqns.(115) and (117) are Eqns of motion

The above Eqns. can be written in matrix form.

$$\begin{bmatrix} m & 0 \\ 0 & J \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} (K_1 + K_2) & -(K_1 a - K_2 b) \\ -(K_1 a - K_2 b) & (K_1 a^2 - K_2 b^2) \end{bmatrix} \begin{Bmatrix} x \\ \theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (118)$$

$$[M]\{\ddot{x}\} + [K]\{x\} = [0]$$

where, $[M]$ -Mass/inertia matrix

$[K]$ - Stiffness matrix

$\{x\}$ -generalized displacement vector

$\{\ddot{x}\}$ -generalized acceleration vector

Stiffness matrix shows that co-ordinate x and θ are dependent on each other. Any change in x reflects in change in θ . As seen from the matrix, the equations of motion are coupled with stiffness. This condition is referred as STATIC COUPLING. coupling in mass matrix is referred as DYNAMIC COUPLING

From the above equation Eqn. (118), it can be seen that system do not have dynamic coupling But, it has static coupling

If the system has a condition $K_1 a = K_2 b$, then, stiffness matrix reduces to:

$$\begin{bmatrix} m & 0 \\ 0 & J \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} (K_1 + K_2) & 0 \\ 0 & (K_1 a^2 - K_2 b^2) \end{bmatrix} \begin{Bmatrix} x \\ \theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (119)$$

Therefore the condition $K_1 a = K_2 b$, uncouples the stiffness matrix.

In Eqn. (119) one can find, the system do not have either Dynamic or Static coupling, the the Eqns. of motion can be written as:

$$m\ddot{x} + (K_1 + K_2)x = 0 \quad (120)$$

Contains only one coordinate, x

$$J\ddot{\theta} + (K_1 a^2 + K_2 b^2)\theta = 0 \quad (121)$$

Contains only one coordinate, θ

Under such conditions, x and θ are independent of each other and are referred as PRINCIPAL COORDINATES.

Solution of uncoupled Eqns. of motion:

The solution of un-coupled eqns of motion is simple, is solving single dof system as shown below. From Eqn.(120).

$$\ddot{x} + \left(\frac{K_1 + K_2}{m} \right) x = 0 \quad (122)$$

$$\omega_1 = \sqrt{\frac{K_1 + K_2}{m}} \text{ rad/s}$$

From Eqn.(121)

$$\ddot{\theta} + \left(\frac{K_1 a^2 + K_2 b^2}{J} \right) \theta = 0 \quad (122)$$

$$\omega_2 = \sqrt{\frac{K_1 a^2 + K_2 b^2}{J}} \text{ rad/s}$$

Example-2

Obtain the equations of motion of the system shown in the Fig.25, check for dynamic and static coupling of co-ordinates. The centre of gravity is away from geometric centre by distance e . The vibration is restricted in plane of paper

m -mass of the system

J -mass MI of the system

G -centre of gravity

C -centre of geometry

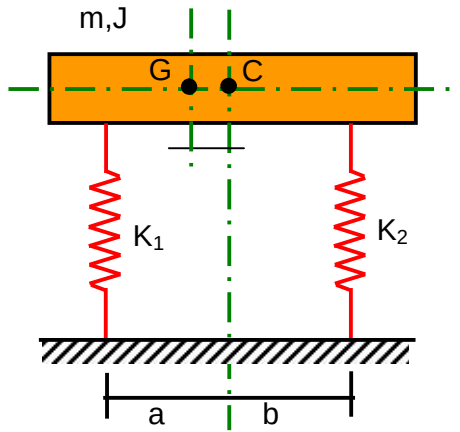


Fig.25. A 2 dof vibratory system having unbalance

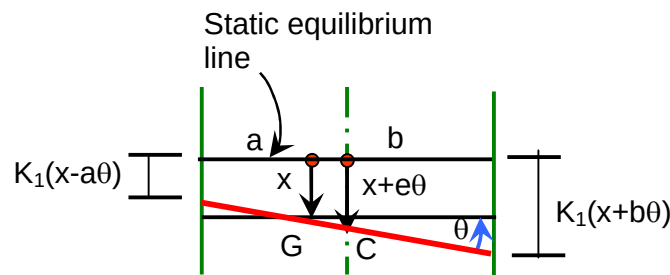


Fig.26 displacements in springs at point 1 and 2 due to vibration.

The system has two generalized co-ordinates, x and θ . x is in Cartesian and θ is in Polar co-ordinate systems. For small disturbance the system the amount displacements in two springs are shown in Fig.26.

Due to some eccentricity e , the changes are:

$$x = x + e\theta$$

$$J = J + me^2$$

Substitute in Eqns. of motion of earlier problem having $e=0$:

Equations of motion for system having $e=0$, (Example-1) are:

$$m\ddot{x} + (K_1 + K_2)x - (K_1a - K_2b)\theta = 0$$

$$J\ddot{\theta} - (K_1a - K_2b)x + (K_1a^2 + K_2b^2)\theta = 0$$

Substitute $x = x + e\theta$ and $J = J + me^2 = J_n$ in above Eqns.

$$m(\ddot{x} + e\ddot{\theta}) + (K_1 + K_2)x - (K_1a - K_2b)\theta = 0$$

$$m\ddot{x} + me\ddot{\theta} + (K_1 + K_2)x - (K_1a - K_2b)\theta = 0 \quad (123)$$

$$J_n\ddot{\theta} - (K_1a - K_2b)x + (K_1a^2 + K_2b^2)\theta = 0 \quad (124)$$

Matrix form of Eqns. of motion is:

$$\begin{bmatrix} m & me \\ 0 & J_n \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} (K_1 + K_2) & -(K_1a - K_2b) \\ -(K_1a - K_2b) & (K_1a^2 + K_2b^2) \end{bmatrix} \begin{Bmatrix} x \\ \theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (125)$$

Eqn.(125) indicates that the system has both dynamic and static coupling. If eccentricity, $e=0$, one can get dynamic uncoupling.

Example-3

Derive expressions for two natural frequencies for small oscillation of pendulum shown in Fig.27 in plane of the paper. Assume rods are rigid and mass less.

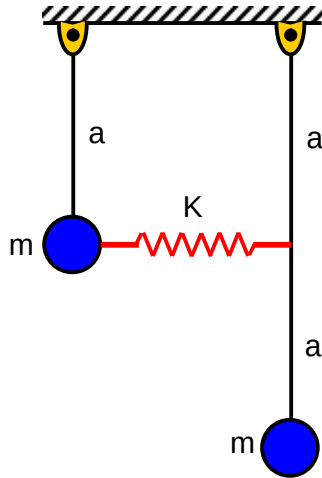


Fig.27. A pendulum system

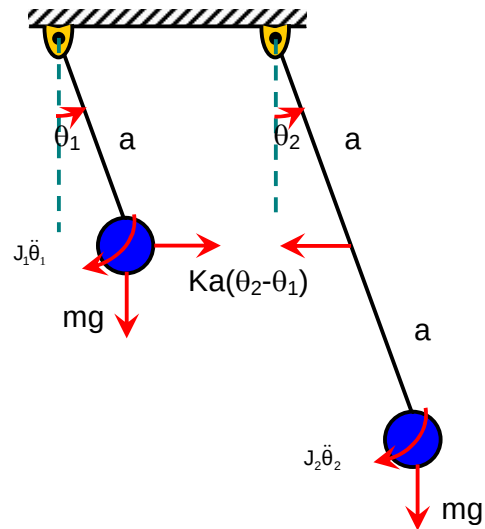


Fig.28. Equilibrium diagram

For small oscillations of the system the equilibrium diagram is shown in Fig.28.

For first mass:

$$J_1 \ddot{\theta}_1 + mg(a \sin \theta_1) - Ka(\theta_2 - \theta_1)(a \cos \theta_1) = 0 \quad (126)$$

as θ is smaller

$$J_1 \ddot{\theta}_1 + mga\theta_1 - Ka^2(\theta_2 - \theta_1) = 0$$

$$(ma^2) \ddot{\theta}_1 + mga\theta_1 - Ka^2\theta_2 + Ka^2\theta_1 = 0$$

The First Eqn. of motion is:

$$ma \ddot{\theta}_1 + (mg + Ka)\theta_1 - Ka\theta_2 = 0 \quad (127)$$

For second mass:

$$J_2 \ddot{\theta}_2 + mg(2a \sin \theta_2) + Ka(\theta_2 - \theta_1)(a \cos \theta_2) = 0 \quad (128)$$

as θ is smaller

$$\{m(2a)^2\} \ddot{\theta}_2 + 2mga\theta_2 + Ka^2\theta_2 - Ka^2\theta_1 = 0$$

Second Eqn. of motion

$$(4ma) \ddot{\theta}_2 - Ka\theta_1 + (2mg + Ka)\theta_2 = 0 \quad (129)$$

Eqns. of motion in matrix form

$$\begin{bmatrix} ma & 0 \\ 0 & 4ma \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} (mg + Ka) & -Ka \\ -Ka & (2mg + Ka) \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (130)$$

For static coupling $Ka=0$, which is not possible.

Solution to governing eqns.:

Assume SHM

$$x_1 = A_1 \sin(\omega t + \phi)$$

$$x_2 = A_2 \sin(\omega t + \phi)$$

The above equations have to satisfy the governing equations of motions Eqns. (127) and (129).

$$[(mg + Ka) - ma\omega^2]A_1 \sin(\omega + \phi) - KaA_2 \sin(\omega + \phi) = 0 \quad (131)$$

$$-KaA_1 \sin(\omega + \phi) - [(2mg + ka) - 4ma\omega^2]A_2 \sin(\omega + \phi) = 0 \quad (132)$$

In above equations $\sin(\omega t + \phi) \neq 0$

Characteristic Eqns. are:

$$[(mg + Ka) - ma\omega^2]A_1 - KaA_2 = 0 \quad (133)$$

$$-KaA_1 - [(2mg + ka) - 4ma\omega^2]A_2 = 0 \quad (134)$$

The above Eqns can be written in determinant form as:

$$\begin{vmatrix} [(mg + Ka) - ma\omega^2] & -Ka \\ -Ka & (2mg + Ka) - 4ma\omega^2 \end{vmatrix} = 0 \quad (135)$$

The above equation is referred as a characteristic determinant, Solving, we get :

$$[(mg + Ka) - ma\omega^2][(2mg + Ka) - 4ma\omega^2] + Ka^2 \quad (136)$$

The above Eqn. is referred as Frequency equation. Solve the frequency Eqn. for Natural frequencies of the system $\omega_1, \dots, \omega_2$.

As the system has two natural frequencies, under certain conditions it may vibrate with first or second frequencies, which are referred as principal modes of vibration. The modes of vibrations can be obtained by the procedure as shown in earlier problems.

Forced Vibrations and Applications

Consider a Two dof system with a forcing function $F_1 = F_0 \sin \omega t$ as shown in Fig.29.

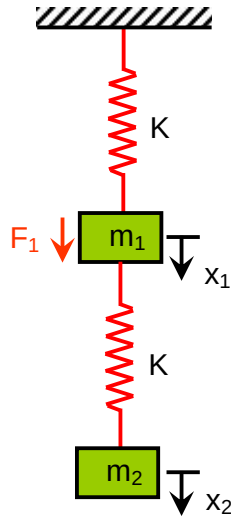


Fig.29 Two dof system with forcing function F_1 on mass 1

The eqns of the motion of the system can be obtained by force balance method. The force equilibrium diagram is shown in Fig.30.

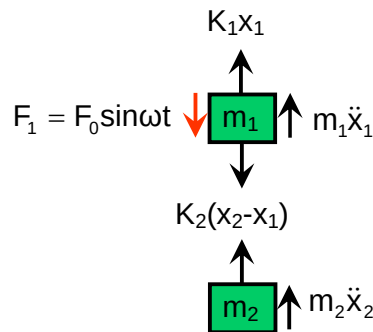


Fig.30 Force equilibrium diagram of the system

From Force equilibrium diagram of mass m_1 :

$$m_1 \ddot{x}_1 + K_1 x_1 - K_2 (x_2 - x_1) = F_0 \sin \omega t$$

$$m_1 \ddot{x}_1 + (K_1 + K_2) x_1 - K_2 x_2 = F_0 \sin \omega t$$

(137)

From Force equilibrium diagram of mass m_2 ;

$$\begin{aligned} m_2 \ddot{x}_2 + K_2(x_2 - x_1) &= 0 \\ m_2 \ddot{x}_2 - K_2 x_1 + K_2 x_2 &= 0 \end{aligned} \quad (138)$$

It is possible to have pure harmonic free vibration for both the masses. Therefore,

Let us assume.

$$x_1 = A_1 \sin \omega t$$

$$x_2 = A_2 \sin \omega t \text{ where } \omega \text{ is the forcing frequency on the system.}$$

Be the forced response of the system, where A_1 and A_2 are the amplitudes to be obtained.

The above equations have to satisfy the governing equations of motions.

$$[(K_1 + K_2) - m_1 \omega^2] A_1 \sin(\omega t + \phi) - K_2 A_2 \sin(\omega t + \phi) = F_0 \sin \omega t \quad (139)$$

$$-K_2 A_1 \sin(\omega t + \phi) - (K_2 - m_2 \omega^2) A_2 \sin(\omega t + \phi) = 0 \quad (140)$$

In above equations,

$$\sin(\omega t + \phi) \neq 0$$

The above equations reduces to: (characteristic equation)

$$[(K_1 + K_2) - m_1 \omega^2] A_1 - K_2 A_2 = F_0 \sin \omega t \quad (141)$$

$$-K_2 A_1 - (K_2 - m_2 \omega^2) A_2 = 0 \quad (142)$$

From above equations obtain A_1 and A_2 by Cramer's rule

$$A_1 = \frac{\begin{vmatrix} F_0 & -K_2 \\ 0 & K_2 - m_2 \omega^2 \end{vmatrix}}{\Delta} \quad (143)$$

$$A_2 = \frac{\begin{vmatrix} (K_1 + K_2) - m_1 \omega^2 & F_0 \\ -K_2 & 0 \end{vmatrix}}{\Delta} \quad (144)$$

where Δ is the determinant of characteristic equations.

$$\Delta = \begin{vmatrix} (K_1 + K_2) - m_1 \omega^2 & -K_2 \\ -K_2 & K_2 - m_2 \omega^2 \end{vmatrix} \neq 0 \quad (145)$$

Solving,

$$\Delta = \{(K_1 + K_2) - m_1 \omega^2\} \{K_2 - m_2 \omega^2\} - K_2^2 \quad (146)$$

If one considers the two vibratory systems separately as shown in Fig.31, the system 1 is a main system and system 2 is an secondary system. This system can be used as Dynamic vibration absorber or Tuned damper by using the amplitude Eqns (143) and (144).

If the system has to be used as Dynamic vibration absorber, the amplitude of vibration of mass m_1 , i.e $A_1=0$.

$$A_1 = \frac{\begin{vmatrix} F_0 & -K_2 \\ 0 & K_2 - m_2\omega^2 \end{vmatrix}}{\Delta} = 0 \quad (147)$$

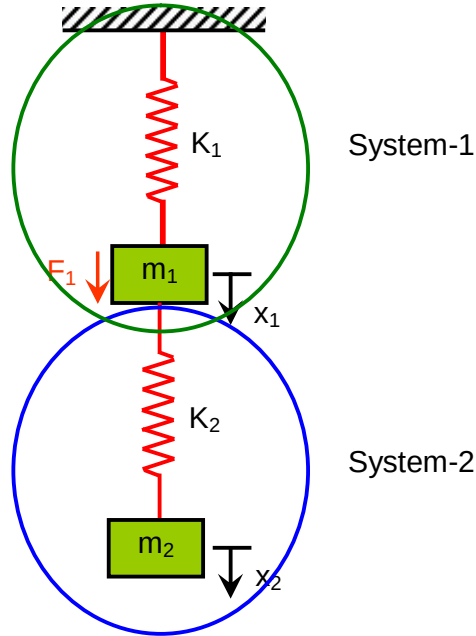


Fig. 31 Concept of vibration absorber

$$\begin{vmatrix} F_0 & -K_2 \\ 0 & K_2 - m_2\omega^2 \end{vmatrix} = 0$$

$$F_0(K_2 - m_2\omega^2) = 0$$

$$K_2 - m_2\omega^2 = 0$$

$$\omega^2 = \frac{K_2}{m_2}$$

$$\omega = \sqrt{\frac{K_2}{m_2}} \text{ rad/s} \quad (148)$$

The above Eqn. is the natural frequency of secondary or absorber system.

Excitation frequency of the system is equal to natural frequency of absorber system.

$$\omega = \sqrt{\frac{K_2}{m_2}} = \omega_2 \quad (149)$$

This shows that if excitation frequency of the system is equal to natural frequency of absorber system, then vibration of the main system is zero.

Undesired maximum vibrations can occur only when main system is under resonance, or near it i.e $\omega_1 = \omega$

So, to reduce undesired vibration of main system mass m_1 , we choose K_2, m_2 in such a way that $\omega_1 = \omega_2$.

$$\sqrt{\frac{K_1}{m_1}} = \sqrt{\frac{K_2}{m_2}} \quad \text{OR} \quad \frac{K_1}{m_1} = \frac{K_2}{m_2} \quad (150)$$

The amplitude A_1 Eqn to be solved to obtain the frequencies,

$$A_1 = \frac{\begin{vmatrix} F_0 & -K_2 \\ 0 & K_2 - m_2 \omega^2 \end{vmatrix}}{\Delta}$$

$$A_1 = \frac{F_0(K_2 - m_2 \omega^2)}{\{(K_1 + K_2) - m_1 \omega^2\}\{K_2 - m_2 \omega^2\} - K_2^2}$$

$$\text{If } \delta_{st} = \frac{F_0}{K_1} \text{ and } \omega_1 = \sqrt{\frac{K_1}{m_1}} \text{ and } \omega_2 = \sqrt{\frac{K_2}{m_2}}$$

Then,

$$\frac{A_1}{\delta_{st}} = \frac{1 - \left(\frac{\omega}{\omega_2}\right)^2}{\left\{\left(1 + \frac{K_2}{K_1}\right) - \left(\frac{\omega}{\omega_1}\right)^2\right\} \left\{1 - \left(\frac{\omega}{\omega_2}\right)^2\right\} - \frac{K_2}{K_1}} \quad (151)$$

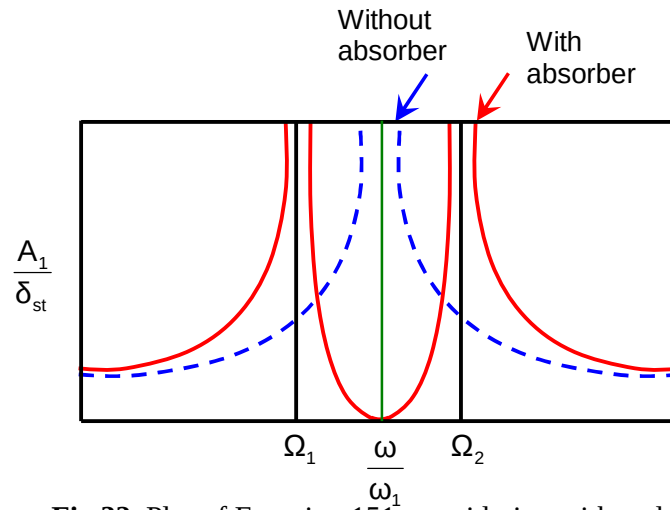


Fig.32. Plot of Equation 151, considering with and without absorber.

To find Ω_1 and Ω_2

If frequency equation is zero, two masses have infinite amplitude of vibrations. The Eqn. is quadratic in ω^2 , therefore there are two values of ω for which amplitude of the system is infinite. Solve for frequency equation Δ .

$$\Delta = \begin{vmatrix} (K_1 + K_2) - m_1\omega^2 & -K_2 \\ -K_2 & K_2 - m_2\omega^2 \end{vmatrix} = 0$$

$$\Delta = \{(K_1 + K_2) - m_1\omega^2\}\{K_2 - m_2\omega^2\} - K_2^2 = 0$$

$$m_1m_2\omega^4 - \{(K_1 + K_2)m_2 + K_2m_1\}\omega^2 + K_1K_2 = 0$$

$$\frac{m_1m_2}{K_1K_2}\omega^4 - \left\{\left(1 + \frac{K_2}{K_1}\right)\frac{m_2}{K_2} + \frac{m_1}{K_1}\right\}\omega^2 + 1 = 0$$

For the absorber system, we have

$$\frac{K_1}{m_1} = \frac{K_2}{m_2}$$

$$\frac{m_2}{m_1} = \frac{K_2}{K_1} = \mu = \frac{\text{mass of absorber system}}{\text{mass of main system}} = \text{Mass ratio} \quad (152)$$

$$\frac{m_1m_2}{K_1K_2}\omega^4 - \left\{\left(1 + \frac{K_2}{K_1}\right)\frac{m_2}{K_2} + \frac{m_1}{K_1}\right\}\omega^2 + 1 = 0$$

$$\frac{m_1m_2}{K_1K_2}\omega^4 = \frac{\omega^4}{\frac{K_1}{m_1} \frac{K_2}{m_2}} = \frac{\omega^4}{\omega_2^4} = r^4$$

$$r^4 - \{(1 + \mu) + 1\} \frac{\omega^2}{\left(\frac{K_2}{m_2}\right)} + 1 = 0$$

$$r^4 - \{2 + \mu\}r^2 + 1 = 0$$

The above Eqn. is quadratic in r^2

$$r_1^2, r_2^2 = \left(\frac{\Omega_{1,2}}{\omega_2}\right)^2 = \left\{1 + \frac{\mu}{2}\right\} \pm \sqrt{\left(\mu + \frac{\mu^2}{4}\right)} \quad (153)$$

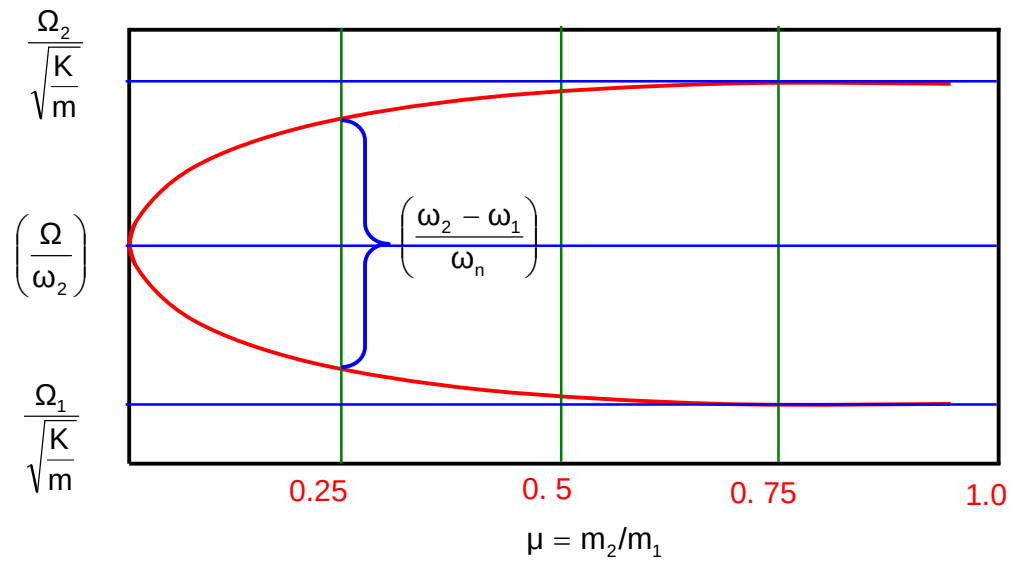


Fig.33. Plot of Equation 153

The Fig.33 indicates that, if mass ratio is greater the spread between the two resonant frequencies is greater. The plot is used to design the absorber system to a main vibratory system to reduce the vibrations of a main system.